

'Recursive' Networks for Conditional Empirical Asset Pricing

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Where we are going

- **Goal:** Develop a tractable machine-learning framework for conditional empirical asset pricing with portfolio rebalancing costs.
- **Broader question:** What can dynamic portfolio choice with transaction costs tell us about key properties of sensible empirical asset pricing models?
- **Approach:**
 - Model portfolio policies incorporating key properties of dynamic optimal portfolio policies.
 - Avoid complete structural solutions requiring generally unavailable state dynamics and quickly becoming computationally intractable.
- **Main building blocks:**

portfolio recursion $\implies$ 'recursive' networks $\implies$ frictional pricing constraints.

Dynamic Portfolio Choice and Recursivity

Which Economic Objective for Empirical Asset-Pricing ML?

Frictional Conditional Asset Pricing

'Recursive' Proximal Network Specifications

Validation and Empirical Research Agenda

Dynamic Portfolio Choice and Recursivity

Recall: Transaction costs make portfolio choice inherently recursive

- In structural multiperiod problems, optimal portfolio decisions are characterized by a **Bellman recursion**:

$$V_t(\mathbf{s}_t) = \max_{\mathbf{a}_t \in \mathcal{A}(\mathbf{s}_t)} \{u_t(\mathbf{s}_t, \mathbf{a}_t) + \mathbb{E}_t[V_{t+1}(\mathbf{s}_{t+1})]\}.$$

- State $\mathbf{s}_t$ summarizes all information relevant for **current decisions** and for determining the **conditional distribution of future investment opportunities**.
- **Without TC**, inherited holdings can typically be **freely re-optimized** and **do not constitute** an additional state variable for the portfolio decision.
- **With TC**, optimal portfolio rebalancing depends on the **inherited portfolio**:

past portfolio choice $\implies$ current portfolio state $\implies$ optimal rebalancing policy.

$\Rightarrow$ TC induce an **endogenous recursive state dependence** in optimal portfolio policies.

Endogenous recursive state dependence

- $\mathbf{x}_{t-}$ denotes the **portfolio inherited** before rebalancing and $\mathbf{x}_t$ the **new portfolio allocation**.
- Portfolio rebalancing costs at time t depend on the **portfolio adjustment**:

$$\Delta \mathbf{x}_t = \mathbf{x}_t - \mathbf{x}_{t-}.$$

⇒ For a given **conditional distribution** of returns $\mathbf{R}_t$ and given return predictors $\mathbf{z}_t$, the optimal portfolio rule still depends on the **inherited holdings**.

- Abstractly, the optimal portfolio policy therefore takes the **recursive form**:

$$\mathbf{x}_t^* = \pi_t(\mathbf{z}_t, \mathbf{c}_t, \mathbf{x}_{t-}), \quad \mathbf{x}_{t-} = \mathcal{T}(\mathbf{x}_{t-1}^*, \mathbf{R}_t),$$

where $\mathcal{T}$ maps the previous optimal holdings into the current pre-rebalancing portfolio, while $\mathbf{c}_t$ collects asset-related cost information.

⇒ This **recursivity** is an **economic implication of TC**, rather than a mere architectural choice.

Transaction costs also generate specific policy geometries

- Convex TC create an incentive to remain close to inherited holdings.
 - With proportional TC, optimal policies feature no-rebalancing regions:
 - Inherited portfolios inside (outside) the region are left unchanged (rebalanced to its boundary);
 - The resulting policy exhibits characteristic nonlinear and non-smooth behavior across rebalancing boundaries.
- ⇒ A frictional policy is not just a conditional target rule $\pi(\mathbf{z}_t)$ determined by current return predictors, but a recursive rebalancing rule $\pi(\mathbf{z}_t, \mathbf{c}_t, \mathbf{x}_{t-})$ around inherited holdings.
- ⇒ These economically structured nonlinearities provide design principles for empirical models of portfolio choice with TC.

What a complete structural portfolio solution would require

- A complete structural characterization of the optimal portfolio policy

$$\mathbf{x}_t^* = \pi(\mathbf{z}_t, \mathbf{c}_t, \mathbf{x}_{t-})$$

requires solving the associated Bellman recursion:

⇒ Repeated conditional expectation computation and optimization over relevant state space.

⇒ Beyond $\mathbf{x}_{t-}$ and $\mathbf{c}_t$, the relevant state $\mathbf{z}_t$ may be high-dimensional and include, e.g.,

$$(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t, \text{predictors of future costs}, \dots).$$

- The Bellman solution must approximate both continuation values and optimal policies, including their no-rebalancing geometry, over the relevant state space.
- This is a computationally formidable task already for structural models with known state dynamics and a relatively small number of assets.

A fully structural approach is even more impractical for empirical asset pricing

- A fully specified transition model for all economically relevant states is **difficult to estimate credibly** and subject to substantial **misspecification risk**.
 - Key **challenges** include:
 - a large and potentially changing **cross-section of assets**;
 - high-dimensional predictive **information sets**;
 - an unknown, time-varying **conditional return distribution**;
 - partly unobserved or imperfectly measured **transaction costs**;
 - ...
- ⇒ Specifying, estimating, and solving the resulting multiperiod dynamic program is therefore **not a practical starting point** for empirical asset pricing.

From structural solutions to empirical policy modelling

- The previous challenges motivate **modelling portfolio policies directly**, rather than specifying, estimating, and solving a complete structural dynamic model.
 - Structural portfolio models nevertheless provide **design principles** for these empirical specifications.
 - In particular, optimal portfolio policies with TC exhibit:
 - dependence on **inherited portfolio holdings**;
 - **recursive dependence** across rebalancing dates;
 - economically structured **nonlinearities and non-smoothness**.
- ⇒ Use structural theory to **discipline the policy class**, rather than to specify and solve the full dynamic model.

Direct recursive policy approximation

- A natural first step is to encode the **recursive structure** directly in the empirical policy:

$$\boxed{\mathbf{x}_t^* = F_{\mathbf{w}}(\mathbf{z}_t, \mathbf{c}_t, \mathbf{x}_{t-})}, \quad \mathbf{x}_{t-} = \mathcal{T}(\mathbf{x}_{t-1}^*, \mathbf{R}_t).$$

- The inherited portfolio $\mathbf{x}_{t-}$ acts as an **endogenous recursive state**, transmitting past portfolio decisions and realized returns to the current policy.
 - The same policy mapping $F_{\mathbf{w}}$, with **fixed parameters $\mathbf{w}$** , is applied across rebalancing dates.
 - Policy inputs evolve through both **new conditioning information** and **inherited holdings**.
- ⇒ Recursion captures the relevant **dynamic dependence**, but does not by itself induce the desired **economic policy geometry**.
- ⇒ This requires an **appropriate economic criterion** for learning the policy parameters $\mathbf{w}$.

Which Economic Objective for Empirical Asset-Pricing ML?

What should an empirical asset-pricing network optimize?

- Direct policy learning requires an **objective criterion** for training the network parameters.
- A standard ML objective targets **conditional return predictability**, e.g.:

$$\hat{\mu}_t = f_w(z_t), \quad \min_w \sum_t \|R_{t+1} - \hat{\mu}_t\|^2.$$

- For **empirical asset pricing**, however, predictability is not the ultimate economic object.
- To give the ML model **economic content**, its objective should be tied to an **optimal portfolio problem** with an associated **pricing-kernel interpretation**.
 - ⇒ The **admissible portfolio class** determines which arbitrages can be exploited and, hence, the relevant pricing restrictions.
 - ⇒ Realistic **asset-pricing restrictions** can hardly abstract from the **TC** faced by investors.

With TC, predictability and economic value can diverge

- Statistical predictability has value only through the **portfolio decisions** exploiting it.
- With TC, exploiting conditional information requires portfolio adjustments whose **economic value** depends on implementation costs:

$$(z_t, c_t, x_{t-}) \longrightarrow \Delta x_t \longrightarrow \underbrace{\text{gross portfolio payoff}}_{\text{investment benefit}} - \underbrace{\sigma_t(\Delta x_t)}_{\text{rebalancing cost}} .$$

- A **statistically successful** predictor may still generate **little economic value**:
 - rapidly changing signals may induce **excessive turnover**;
 - small predictive gains may be **dominated by rebalancing costs**.
 - Predictive accuracy and economic value may be **potentially competing objectives**.

⇒ The learning criterion should be built on the **portfolio opportunities generated net of TC**.

Learn portfolio policies through their economic objective

- Replace the two-step approach with a **direct optimization** of conditional portfolio criterion:

$$\max_w \mathbb{E}_t [\mathcal{U}(\mathbf{x}'_{w,t} \mathbf{R}_{t+1}, \sigma_t(\mathbf{x}_{w,t} - \mathbf{x}_{w,t-}))],$$

where $\mathbf{x}_{w,t}$ is parameterized by a **recursive policy network**.

⇒ Conditional information is useful only if it generates **valuable portfolios after TC**.

- For empirical asset pricing with TC, however, **not all portfolio objectives are equally useful**.

⇒ We want an objective that is simultaneously **tractable under TC** and has a **transparent mapping into asset-pricing restrictions**.

⇒ Which portfolio criterion delivers both?

Frictional Conditional Asset Pricing

From frictional portfolio choice to frictional asset pricing

- We seek a tractable conditional portfolio problem with TC that admits a transparent dual asset-pricing interpretation.

⇒ We can characterize asset-pricing restrictions through the best portfolio opportunities available to investors subject to TC.

- With TC, these opportunities depend not only on current information and conditional return distributions, but also on investors' pre-trade portfolio holdings:

⇒ The resulting portfolio problem – and its associated pricing restrictions – are inherently conditional and recursive.

⇒ This structure motivates both the economic learning objective and the recursive network specification developed below.

Conditional asset pricing with TC

- Consider N excess returns R_{t+1} and a inherited portfolio θ_{t-} at date t before rebalancing.
- Rebalancing from θ_{t-} to θ_t incurs a positive convex TC:

$$\sigma_t(\theta_t - \theta_{t-}), \quad \sigma_t(0) = 0.$$

- Trading frictions restrict both the portfolio opportunities available to investors and the conditional asset valuations consistent with no-arbitrage:

⇒ We formally exploit this link to formulate a portfolio-learning objective with direct frictional asset-pricing interpretation.

TC imply convex pricing restrictions

- In a **frictional arbitrage-free economy**, any admissible Stochastic Discount Factor (SDF) satisfies

$$\mathbb{E}_t[M_{t+1}R_{t+1}] \in \mathcal{C}_t.$$

- The set of admissible valuations is **determined locally** by the transaction-cost function (and vice-versa):

$$\mathcal{C}_t = \partial\sigma_t(0) = \{\eta_t : \eta_t^\top \theta \leq \sigma_t(\theta), \forall \theta\}.$$

- TC do more than **penalize portfolio turnover**:

⇒ They simultaneously determine the **conditional excess-return valuations** compatible with **no-arbitrage**.

⇒ **Non-differentiable** cost components generate **non-singleton valuation sets** $\mathcal{C}_t$.

A tractable conditional portfolio objective

- Consider the frictional conditional **maximum-Sharpe** portfolio problem:

$$\min_{\theta_t} \left\{ \frac{1}{2} \mathbb{V}_t(\theta_t^\top \mathbf{R}_{t+1}) - \mathbb{E}_t[\theta_t^\top \mathbf{R}_{t+1}] + \sigma_t(\theta_t - \theta_{t-}) \right\}.$$

- Let

$$\mu_t = \mathbb{E}_t[\mathbf{R}_{t+1}], \quad \Sigma_t = \mathbb{V}_t(\mathbf{R}_{t+1}), \quad \theta_t^{SR} = \Sigma_t^{-1} \mu_t.$$

Then, up to a term independent of θ_t , the problem is equivalent to

$$\min_{\theta_t} \left\{ \frac{1}{2} \|\theta_t - \theta_t^{SR}\|_{\Sigma_t}^2 + \sigma_t(\theta_t - \theta_{t-}) \right\}.$$

⇒ Optimal rebalancing trades off distance from the **frictionless conditional target** against the **cost of moving from inherited holdings**.

Dual interpretation as minimal state-price adjustment

- The optimal portfolio induces a tradable linear SDF

$$M_{t,t+1} = 1 - \theta_t^\top (\mathbf{R}_{t+1} - \mathbb{E}_t[\mathbf{R}_{t+1}]) .$$

- For **sublinear costs**, the corresponding dual problem takes the form

$$\min_{M_{t+1} \in \mathcal{M}_t} \mathbb{E}_t \left[\frac{1}{2} (M_{t+1} - M_{t-,t+1})^2 \right] ,$$

where

$$\mathcal{M}_t = \{M : \mathbb{E}_t[M] = 1, \quad \mathbb{E}_t[M\mathbf{R}_{t+1}] \in \mathcal{C}_t\} .$$

⇒ Rebalancing implements the **smallest state-price adjustment** required to **restore** the current frictional pricing restrictions.

⇒ Portfolio and pricing problems provide **primal and dual representations of the same adjustment**.

Projection properties and no-rebalancing region

- Under **sublinear costs**, the optimal portfolio admits the projection representation

$$\theta_t^0 = P_{\mathcal{C}_t^P}^{\Sigma_t}(\theta_{t-}), \quad \mathcal{C}_t^P = \{\theta_t^{SR}\} - \Sigma_t^{-1}\mathcal{C}_t.$$

- If $\theta_{t-} \in \mathcal{C}_t^P$, the inherited portfolio already satisfies the frictional optimum:

⇒ **No rebalancing** and **no state-price adjustment** are required.

- Otherwise, θ_{t-} is projected onto $\mathcal{C}_t^P$ under the **conditional covariance metric**:

⇒ TC generate a state-dependent **no-rebalancing region** and a **convex adjustment operator** outside it.

⇒ This provides a **theoretical template** for the **recursive policy operator** to be approximated by the network.

What structure should the network inherit?

1. **Recurrence**: current decision depends on inherited optimal holdings after return updating,

$$\theta_t^0 = \Phi(\mathcal{I}_t^R, \sigma_t, \theta_{t-1}^0 \odot \mathbf{R}_t^g, \gamma_{t-1}^0).$$

2. **Adversarial structure** under **sublinear costs**:

$$\sigma_t(\Delta\theta_t) = \sup_{\eta_t \in \mathcal{C}_t} \Delta\theta_t^\top \eta_t,$$

⇒ Links **portfolio adjustments** to the **worst-case admissible asset valuation**.

3. **Proximal structure**:

⇒ The portfolio update is obtained as a **convex minimum-distance adjustment** of inherited holdings toward the no trade region $\mathcal{C}_t^P$.

'Recursive' Proximal Network Specifications

Baseline empirical specification: learn the recursive policy directly

- Use a shared policy network

$$(\varphi_t, \zeta_t) = \Phi_{\mathbf{w}}(\mathcal{I}_t^R, \sigma_t, \varphi_{t-1} \odot \mathbf{R}_t^g, \zeta_{t-1}).$$

- Estimate it through the aggregated (unconditional) frictional objective

$$\min_{\mathbf{w}} \mathbb{E} \left[\frac{1}{2} (-\varphi_t^\top \mathbf{R}_{t+1} - \zeta_t)^2 + \zeta_t + \sigma'_t(0, \varphi_t - \varphi_{t-1} \odot \mathbf{R}_t^g) \right].$$

- The network **learns the policy mapping** while the **dual portfolio** loss embeds the corresponding **frictional asset-pricing restrictions**.

Why this is empirically attractive

- It **avoids explicitly estimating** a separate conditional model for every object entering the structural solution.
- The same network can absorb **nonlinear interactions** among:
 - return-predictive characteristics and macro variables;
 - current cost signals;
 - inherited portfolio positions;
 - latent information summarized recursively from the past.
- Importantly, the **unconditional training problem** exploits a large **time-series/cross-sectional sample** while targeting a **conditional policy**.

A remaining obstacle: the conditional covariance matrix

- The exact projection geometry depends on

$$\Sigma_t = \mathbb{V}_t(\mathbf{R}_{t+1}) \in \mathbb{R}^{N \times N}.$$

- In a large cross-section this is a high-dimensional conditional object with $O(N^2)$ entries.
- Estimation error in Σ_t^{-1} directly affects:
 - the frictionless max-Sharpe ratio target;
 - the location, scale and rotation of the no-rebalancing region;
 - the metric in which inherited holdings are projected.
- This motivates policies that preserve convex adjustment structure **without requiring a plug-in estimate of the full conditional covariance matrix!**

Theory-guided restriction: a proximal network policy

- Replace the unrestricted output map by a family of **proximal operators**:

$$\varphi_t = \arg \min_{\theta} \left\{ g_{\xi}(\theta; \mathcal{I}_t) + \frac{1}{2} \|\theta - \varphi_{t-1} \odot \mathbf{R}_t^g\|_2^2 \right\}.$$

- $g_{\xi}(\cdot; \mathcal{I}_t)$ is a convex penalty in the portfolio argument represented as a **Input Convex Neural Network**.
- This **structurally regularizes** the policy class by replacing the unknown Σ_t -induced adjustment geometry with a tractable **surrogate proximal geometry**.
- The network learns the conditional portfolio adjustment, **exploring the unknown trade-off** between investment opportunities, risk, and rebalancing costs.

Beyond Euclidean proximity: learned conditional geometry

- The Euclidean proximal map avoids direct covariance estimation but treats all portfolio directions symmetrically.
- A natural extension is a Bregman-proximal policy:

$$\varphi_t = \arg \min_{\theta} \{g_{\xi}(\theta; \mathcal{I}_t) + D_{\psi_{\omega}}(\theta, \varphi_{t-1} \odot \mathbf{R}_t^g; \mathcal{I}_t)\}.$$

- The curvature of ψ_{ω} provides a learned, state-dependent geometry that can help approximate economically relevant risk directions.

⇒ This suggests a progression from learned policy to learned convex policy geometry.

Cost uncertainty fits naturally into the recursive framework

- Actual implementation costs may only be observed **after** the portfolio is formed.
- If σ_{t+} is the realized **future cost**,

$$\sigma_t(\Delta\theta_t) = \mathbb{E}_t[\sigma_{t+}(\Delta\theta_t)].$$

- The policy can condition on **distinct information sets** for returns and costs:

$$(\varphi_t, \zeta_t) = \Phi_{\mathbf{w}}(\mathcal{I}_t^R, \mathcal{I}_t^C, \varphi_{t-1} \odot \mathbf{R}_t^g, \zeta_{t-1}).$$

- This raises an important empirical question unavailable under a frictionless modelling framework:

How valuable is cost predictability relative to return predictability?

Validation and Empirical Research Agenda

What should we validate before going to large-scale data?

- Use **controlled simulations** where true conditional moments generate an infeasible **Oracle policy** with TC.
- Ask whether learned recursive policies reproduce:
 - Oracle portfolio returns and portfolio directions;
 - conditional Sharpe ratios;
 - reduced trading when costs rise;
 - no-rebalancing behavior and state-dependent adjustment;
 - persistence of portfolios and state prices;
 - the changing geometry induced by correlations and factor risk.
- Compare **unrestricted 'recursive' networks** with **theory-guided** proximal/ICNN policies.

Preliminary simulation evidence from the draft

- The baseline deep-learning policy can closely track the Oracle in the controlled design; without costs it captures about 94.8% of the Oracle Sharpe ratio.
- In frictional simulations, the proximal ICNN architecture improves alignment with the Oracle relative to the baseline network:

	Baseline DL	ICNN / proximal
Return correlation with Oracle	0.68	0.75
Mean conditional Sharpe ratio	0.73	0.78
Sharpe-ratio drop vs. Oracle	29.1%	24.1%

Tentative message: structural regularization appears especially useful precisely when frictions make the policy geometry nontrivial.

Conclusion: From recursive structures to empirical conditional asset pricing

- **First principle:** transaction costs make portfolio/SDF adjustment inherently recursive because inherited holdings matter.
- **Empirical strategy:** learn the recursive conditional policy directly rather than solve a fully specified multiperiod structural model.
- **Asset-pricing discipline:** use the dual frictional max-Sharpe problem to connect portfolio learning to admissible SDF valuations.
- **Theory-guided architecture:** progressively impose proximal/convex structure to regularize high-dimensional empirical policies.
- Prospective large-scale application: many assets, rich return and cost predictors, and explicit comparison of the **economic value of predicting returns versus trading costs**.